

Manas Pandey

SUMMARY

Automotive Embedded Software Engineer with 5+ years of experience in Embedded C development, AUTOSAR (BSW, MCAL, RTE), and ECU integration. Strong expertise in automotive communication protocols (CAN, LIN, Ethernet) and UDS diagnostics (ISO 14229). Hands-on experience with CAPL scripting, CANoe/CANalyzer tools, and system-level debugging. Proven track record in automotive domain projects (instrument cluster / ECU systems) with strong problem-solving and debugging skills. Open to immediate opportunities and relocation to Bengaluru.

EXPERIENCE

Software Engineer II

Visteon Corporation

April 2023 – Present, Pune, India

- Developed embedded software using Embedded C for automotive ECUs
- Worked on AUTOSAR architecture, including BSW, MCAL, and RTE configuration and integration
- Configured and validated CAN, LIN, and Ethernet communication stacks
- Implemented and tested UDS diagnostic services (DCM, DEM)
- Developed and used CAPL scripts in CANoe for simulation and testing
- Performed ECU integration, system-level debugging, and validation
- Debugged communication issues using CANoe, CANalyzer, and TRACE32
- Collaborated in Agile environment with cross-functional teams.

Software Engineer I

Visteon Corporation

Jan 2022 – March 2023, Chennai, India

- Worked on AUTOSAR BSW modules, MCAL drivers, and RTE configuration
- Implemented and validated CAN communication stack and UDS diagnostics
- Developed CAPL scripts for test automation and validation
- Debugged issues related to communication, memory, and timing
- Supported ECU integration and testing activities
- Followed ASPICE-compliant development processes.

Junior Software Engineer

Visteon Corporation

Jan 2021 – Dec 2021, Chennai, India

- Developed embedded applications using C programming
- Implemented low-level drivers and supported peripheral integration
- Supported CAN/LIN communication testing and diagnostics
- Assisted in AUTOSAR configuration and ECU integration
- Performed debugging using JTAG tools.

PROJECT

AUTOSAR ECU Development & Integration

- Configured and integrated BSW, MCAL, and RTE layers
- Performed ECU integration, validation, and debugging.

CAN Communication & Diagnostics

- Implemented CAN stack and UDS diagnostic services (ISO 14229)
- Developed CAPL scripts for simulation, testing, and validation
- Debugged communication issues using CANoe/CANalyzer.

Automotive Embedded Systems (Cluster/ECU)

- Developed software for automotive ECU and instrument cluster systems
- Ensured real-time data communication and system stability.

EDUCATION

M.Tech, Embedded Systems

BITS Pilani • 2024–2026

- In Progress.

SKILLS

Embedded C, AUTOSAR: BSW, MCAL, RTE
Protocols: CAN, LIN, Ethernet, UDS
Tools: CANoe, CANalyzer, CAPL, TRACE32, JTAG
Concepts: ECU Integration, Debugging, RTOS
Process: ASPICE, Agile/Scrum
Version Control: Git, SVN
